

All-Prime Borel $j_!$ Solid Dual

Autonomous discussion Pass 94 computes the Verdier/solid dual of the all-prime Borel $j_!$ coefficient.

Pass 93 identified the all-prime coefficient as

$$\mathfrak{B}_{j_!}^{\text{cont}} = \mathbb{Q}^\times \ltimes \varprojlim_{S \in \mathbb{P}} j_{S,!} \mathcal{V}_S,$$

whose unipotent cohomology is

$$\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}.$$

Solid duality gives

$$D\epsilon \simeq \mathbb{Q}[-1].$$

Therefore the dual unipotent is a shifted boundary object, not an opposite degree-0 unipotent group. Keeping the global Levi, the dual Borel shadow is a Levi-marked boundary object with unipotent part $\mathbb{Q}[-1]$ and contragredient $\mathbb{Q}^\times$ action.

The finite signed Verdier rule survives. For the finite recollement boundary

$$d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1},$$

duality sends

$$D(d_S) = -d_S^T, \quad D^2(d_S) = d_S.$$

The sign is visible over $\mathbb{Z}$ and invisible modulo 2.

All-prime, the same sign appears as the antipode on the bidual:

$$\eta_\epsilon = -\text{id}_\epsilon.$$

It is a functional-equation shadow only in the boundary sense. The surviving object is the finite-adele extension

$$0 \rightarrow \mathbb{Q} \rightarrow \mathbb{A}_f \rightarrow \epsilon \rightarrow 0,$$

or equivalently the degree-1 class in $\text{Ext}_{\text{Solid}}^1(\epsilon, \mathbb{Q})$.

The no-Weyl-flip wall remains:

$$\text{Hom}_{\text{Solid}}^0(\epsilon, \mathbb{Q}) = 0.$$

Thus Pass 94 does not create a standard intertwiner, opposite unipotent, or degree-0 Fourier/Weyl operator. It records a boundary-only functional equation, compatible with the Pass-81 no-functional-equation wall.

The finite checker `artifacts/reports/pass94-all-prime-borel-jshriek-solid-dual-check.json` verifies the signed transpose rule, duality squared, support-dual behavior, the solid identity $D\epsilon = \mathbb{Q}[-1]$, the degree-1 finite-adele boundary, and the absence of a degree-0 Weyl flip.